

Formal Proofs for the Edge-Cap Encoding Scheme

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1 Correctness and Concatenation Safety

This document establishes the formal correctness of the edge-cap construction through two results. Lemma 1 guarantees that the cap selection procedure always succeeds — that is, for every core codeword produced by the DP generator, a valid cap pair satisfying all three conditions always exists. Theorem 1 then guarantees that the direct concatenation of any two capped codewords satisfies the homopolymer constraint $r \leq 3$ at every position, including the junction. Corollary 1 extends this to arbitrary-length concatenations.

We use the following notation throughout. Let $\Sigma = \{\mathbf{A}, \mathbf{C}, \mathbf{G}, \mathbf{T}\}$ be the DNA alphabet, $\Sigma_{\text{GC}} = \{\mathbf{G}, \mathbf{C}\}$, and $\Sigma_{\text{AT}} = \{\mathbf{A}, \mathbf{T}\}$. For a core codeword $w = w_1 w_2 \dots w_n \in \mathcal{C}_n$, let $g_{\text{core}} = |\{i : w_i \in \Sigma_{\text{GC}}\}|$ denote its GC base count. The admissible GC counts for a sequence of length m are:

$$n_{\min}(m) = \lceil 0.4m \rceil, \quad n_{\max}(m) = \lfloor 0.6m \rfloor. \quad (1)$$

The capped codeword is $\hat{w} = \ell \cdot w \cdot r$ of length $n + 2$, where $\ell, r \in \Sigma$ are the left and right caps respectively.

1.1 Lemma 1: Cap Existence

Lemma 1 (Cap Existence). *For every core codeword $w \in \mathcal{C}_n$, the cap selection procedure always finds a valid pair $(\ell, r) \in \Sigma^2$ satisfying:*

1. $\ell \neq w_1$ (left cap differs from first core base),
2. $r \neq w_n$ (right cap differs from last core base),
3. $\text{GC}(\ell \cdot w \cdot r) \in [0.4, 0.6]$ (capped word satisfies GC-content constraint),

and never raises a failure.

Proof. Since $w \in \mathcal{C}_n$, the core satisfies the GC-content constraint:

$$n_{\min}(n) \leq g_{\text{core}} \leq n_{\max}(n), \quad \text{i.e.,} \quad \lceil 0.4n \rceil \leq g_{\text{core}} \leq \lfloor 0.6n \rfloor. \quad (2)$$

For the capped word of length $n + 2$, the required GC count satisfies:

$$n_{\min}^{\text{cap}} = \lceil 0.4(n + 2) \rceil, \quad n_{\max}^{\text{cap}} = \lfloor 0.6(n + 2) \rfloor. \quad (3)$$

A cap pair (ℓ, r) is GC-valid if and only if the total GC count $g = g_{\text{core}} + \mathbf{1}[\ell \in \Sigma_{\text{GC}}] + \mathbf{1}[r \in \Sigma_{\text{GC}}]$ satisfies $n_{\min}^{\text{cap}} \leq g \leq n_{\max}^{\text{cap}}$. Since each cap contributes either 0 or 1 to g , the set of

achievable values of g given g_{core} is $\{g_{\text{core}}, g_{\text{core}} + 1, g_{\text{core}} + 2\}$, corresponding to cap pairs with 0, 1, or 2 GC bases respectively. We show that at least one of these three values always falls within $[n_{\text{min}}^{\text{cap}}, n_{\text{max}}^{\text{cap}}]$ by considering three exhaustive cases.

Case 1: $g_{\text{core}} \leq n_{\text{max}}^{\text{cap}} - 2$. Then $g_{\text{core}} + 2 \leq n_{\text{max}}^{\text{cap}}$. It remains to verify that $g_{\text{core}} + 2 \geq n_{\text{min}}^{\text{cap}}$, i.e. $g_{\text{core}} \geq n_{\text{min}}^{\text{cap}} - 2$. From (2):

$$g_{\text{core}} \geq \lceil 0.4n \rceil \geq \lceil 0.4(n+2) \rceil - 1 = n_{\text{min}}^{\text{cap}} - 1 > n_{\text{min}}^{\text{cap}} - 2. \quad (4)$$

Hence $g = g_{\text{core}} + 2$ is GC-valid, achieved by choosing both caps from Σ_{GC} .

Case 2: $g_{\text{core}} \geq n_{\text{min}}^{\text{cap}}$. Then $g_{\text{core}} \geq n_{\text{min}}^{\text{cap}}$. It remains to verify that $g_{\text{core}} \leq n_{\text{max}}^{\text{cap}}$. From (2):

$$g_{\text{core}} \leq \lfloor 0.6n \rfloor \leq \lfloor 0.6(n+2) \rfloor = n_{\text{max}}^{\text{cap}}. \quad (5)$$

Hence $g = g_{\text{core}}$ is GC-valid, achieved by choosing both caps from Σ_{AT} .

Case 3: $n_{\text{max}}^{\text{cap}} - 1 \leq g_{\text{core}} \leq n_{\text{min}}^{\text{cap}} - 1$. In this case $g_{\text{core}} + 1$ lies in $[n_{\text{min}}^{\text{cap}}, n_{\text{max}}^{\text{cap}}]$, achieved by choosing one cap from Σ_{GC} and one from Σ_{AT} .

It remains to verify that the homopolymer conditions $\ell \neq w_1$ and $r \neq w_n$ do not prevent finding a pair with the required GC contribution in each case.

Case 1 requires both $\ell, r \in \Sigma_{\text{GC}} = \{\mathbf{G}, \mathbf{C}\}$. For any $w_1 \in \Sigma$, at most one element of Σ_{GC} equals w_1 , so at least one valid choice for ℓ remains in Σ_{GC} . The same argument applies independently to r with respect to w_n . Since the choices of ℓ and r are independent, a valid pair always exists.

Case 2 requires both $\ell, r \in \Sigma_{\text{AT}} = \{\mathbf{A}, \mathbf{T}\}$. By the same argument with $|\Sigma_{\text{AT}}| = 2$, for any $w_1, w_n \in \Sigma$ at least one valid choice remains in each class independently.

Case 3 requires one cap from Σ_{GC} and one from Σ_{AT} . Since each content class has two elements and the homopolymer condition eliminates at most one element per class per boundary base, at least one valid choice remains in each class. The assignment of which class goes to ℓ and which to r can be swapped if needed, providing two candidate pairs.

In all cases a valid pair (ℓ, r) satisfying both the GC-content and homopolymer conditions exists, so the cap selection procedure always terminates successfully. $\square$

1.2 Theorem 1: Concatenation Safety

Theorem 1 (Concatenation Safety). *Let $\hat{w}^{(1)}$ and $\hat{w}^{(2)}$ be any two capped codewords produced by the edge-cap construction. Their direct concatenation $\hat{w}^{(1)}\hat{w}^{(2)}$ satisfies the homopolymer constraint $r \leq 3$ at every position.*

Proof. We verify the constraint at three regions of the concatenated sequence.

Interior of $\hat{w}^{(1)}$. The core $w^{(1)}$ satisfies $r \leq 3$ by construction, as it is a member of $\mathcal{C}_n$. The left cap $\ell^{(1)}$ satisfies $\ell^{(1)} \neq w_1^{(1)}$, so no run extends from the core into the left cap. The right cap $r^{(1)}$ satisfies $r^{(1)} \neq w_n^{(1)}$, so no run extends from the core into the right cap. Hence $\hat{w}^{(1)}$ individually satisfies $r \leq 3$.

Interior of $\hat{w}^{(2)}$. By the same argument, $\hat{w}^{(2)}$ individually satisfies $r \leq 3$.

Junction. The junction between $\hat{w}^{(1)}$ and $\hat{w}^{(2)}$ comprises the subsequence $\dots w_n^{(1)} r^{(1)} \ell^{(2)} w_1^{(2)} \dots$. We consider the maximum possible run length at this junction:

- Since $r^{(1)} \neq w_n^{(1)}$, the run ending at $r^{(1)}$ has length at most 1 extending leftward from $r^{(1)}$.
- Since $\ell^{(2)} \neq w_1^{(2)}$, the run beginning at $\ell^{(2)}$ has length at most 1 extending rightward from $\ell^{(2)}$.
- The only way a run of length 3 or more could form at the junction is if $r^{(1)} = \ell^{(2)}$ and this base also equals $w_n^{(1)}$ or $w_1^{(2)}$. But $r^{(1)} \neq w_n^{(1)}$ and $\ell^{(2)} \neq w_1^{(2)}$ by construction, so any run at the junction is confined to at most the two bases $r^{(1)}$ and $\ell^{(2)}$, giving a maximum junction run length of 2.

Since $2 \leq 3$, the homopolymer constraint is satisfied at the junction. Combining all three regions, $\hat{w}^{(1)}\hat{w}^{(2)}$ satisfies $r \leq 3$ at every position. $\square$

1.3 Corollary 1: Arbitrary-Length Concatenation

Corollary 1 (Arbitrary-Length Concatenation). *Any finite concatenation $\hat{w}^{(1)}\hat{w}^{(2)} \dots \hat{w}^{(K)}$ of Edge-Cap codewords satisfies $r \leq 3$ at every position. No knowledge of neighbouring codewords is required during encoding or decoding.*

Proof. The result follows by applying Theorem 1 to each consecutive pair $(\hat{w}^{(i)}, \hat{w}^{(i+1)})$ for $i = 1, \dots, K-1$. Since each junction independently satisfies $r \leq 3$ and each codeword interior independently satisfies $r \leq 3$, the full concatenation satisfies the constraint everywhere. Because the safety guarantee at each junction depends only on the cap bases of the two adjacent codewords — which are determined entirely from the core codewords themselves — and not on any global encoding state, the encoder and decoder may process all K codewords in any order, including fully in parallel. $\square$